

PLamb_Test_10V6c_plus — Program Guide

Version v1.1 — Updated 13 Nov 2025 (AEST)

Executive Summary

This v1.1 update documents a staged DESI DR2 BAO-only workflow for FR with CPL dark-energy parameterization, clarifies recommended MCMC settings to avoid NaN autocorrelation times, and adds guidance on Noether/link penalties for low-N data.

We recommend:

- (i) stabilise on constant-w (w_0) with BAO-only,
- (ii) upgrade to CPL (w_0, w_a) with broader start scatter and correlated moves, and
- (iii) add SN/Planck/CC before enabling extra FR axes ($A_{\text{acc}}, n_{\text{acc}}, \text{delta}, \text{eta}$).

Data & Paths

Ensure the following files and constants are set:

- DESI DR2 BAO CSV: /Users/boyde/Downloads/DESI_DR2_BAO.csv
- DESI DR2 BAO covariance: /Users/boyde/Downloads/DESI_DR2_BAO_cov.txt
- Sound horizon (rd): 147.09 Mpc (set via --rd 147.09)

Quick-Start Recipes (Staged Workflow)

Run these in sequence.

Stage A provides a stable baseline (constant-w).

Stage B upgrades to CPL with settings that encourage healthy mixing.

Stage C adds additional datasets and, only then, optional FR extensions.

Stage A — Stabilise with constant-w (BAO-only)

```
%run "/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py" \  
  --out "plamb_runs/desi_tests/FR_w0_softlink_c0_desionly" \  
  --model FR --flat --de-model w0 \  
  --bao "/Users/boyde/Downloads/DESI_DR2_BAO.csv" \  
  --bao-cov "/Users/boyde/Downloads/DESI_DR2_BAO_cov.txt" \  
  --rd 147.09 \  
  --bao-c-mode c0 --c-model loglpz \  
  --link-soft --kappa-em-gc 1.0 --link-sigma 0.25 \  
  --noether combo_soft --noether-strength 1.0 \  
  --sampler emcee --nwalkers 128 --nsteps 12000 --nburn 3000 \  

```

```
--preopt --preopt-maxiter 400 --preopt-ftol 1e-6 \
--seed-near-best --seed-width-frac 0.30 \
--tight-priors \
--corner --save-chain --save-figs --save-args-json --verbose
```

Rationale: constant-w mixes well on DESI-only and gives a stable baseline for w0.

Stage B — Upgrade to CPL (BAO-only, well-mixed)

```
%run "/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py" \
--out "plamb_runs/desi_tests/FR_CPL_softlink_c0_desionly_v2" \
--model FR --flat --de-model cpl \
--bao "/Users/boyde/Downloads/DESI_DR2_BAO.csv" \
--bao-cov "/Users/boyde/Downloads/DESI_DR2_BAO_cov.txt" \
--rd 147.09 \
--bao-c-mode c0 --c-model loglpz \
--link-soft --kappa-em-gc 1.0 --link-sigma 0.25 \
--noether combo_soft --noether-strength 1.5 \
--sampler emcee --nwalkers 128 --nsteps 12000 --nburn 3000 \
--preopt --preopt-maxiter 400 --preopt-ftol 1e-6 \
--seed-near-best --seed-width-frac 0.30 \
--tight-priors \
--corner --save-chain --save-figs --save-args-json --verbose
```

Why these settings: broader starts (seed-width-frac ~0.30) plus allowing correlated moves improves mixing and avoids NaN autocorrelation seen with tight seeds and KDE disabled. Reduce Noether strength so the physical BAO fit dominates at this stage.

Stage C — Add constraining datasets (then re-enable FR extras)

After Stage B mixes, add SN and/or Planck (and later CC).

Only after that, consider FR extras (A_acc, n_acc) or delta/eta.

Example: append Pantheon+SH0ES and/or Planck distance priors to the Stage B command, then evaluate fits and re-tune penalties as needed.

Troubleshooting & Diagnostics (v1.1)

- Symptom: IAT (τ) shows NaN for parameters (often w0, wa, delta, eta).

Fix: Increase --seed-width-frac to ~0.25–0.35; do not use --no-kde-move; avoid enabling weakly constrained axes (e.g., --delta-param) on BAO-only.

- Symptom: Acceptance ≈ 0.18 looks fine but chains show limited wander.

Fix: Verify starting scatter; reduce penalty dominance; check corner plots for narrow spikes.

- Target: Aim for steps_post/ $\tau_{\max} \gtrsim 20$. Extend steps or simplify parameter set if unmet.
- Data vs parameter count: With 13 BAO points, keep active parameters modest (≈ 4 –5) before adding SN/Planck/CC.

Noether / Link Penalties — Practical Settings

For BAO-only diagnostics, set `--noether-strength` between 1.0 and 1.5 to avoid penalty-dominated posteriors.

Record both reported χ^2 and penalised χ^2 ; if penalty contributes >30–40% of the total at DESI-only stage, lower the strength so the physical BAO fit (χ^2_{BAO}) is the driver.

Figures Included (1V1)

- `corner_FR_w0_softlink_c0_desionly.png` — Baseline constant-w posterior.
- `corner_FR_CPL_softlink_c0_desionly_v2.png` — CPL posterior after stabilised run.
- `trace_FR_CPL_softlink_c0_desionly_v2.png` — Walker traces (`w0`, `wa` should show healthy wander).
- `bestfit_FR_*_bao_only.png` — BAO residuals vs model (caption: dataset & `r_d`).
- `penalty_breakdown_FR_*_bao_only.png` — χ^2 components (BAO, link, noether).

Add a one-line caption under each figure noting dataset, active parameters, and whether penalties are included in the plotted χ^2 .

Known Results (DESI-only, FR+CPL) — Observations

In recent DESI-only runs, χ^2_{BAO} remained essentially identical (~ 9.159) for `--bao-c-mode c0` versus `c_of_z` with FR+CPL and the specified penalties, indicating low sensitivity of the BAO-only fit to this toggle under these settings.

Avoid over-interpreting `w0`, `wa`, `A_acc`, `n_acc` from BAO-only; add SN/Planck for meaningful constraints.

Recommended Defaults for BAO-only (v1.1)

Flag	v1.1 default for BAO-only	Notes
<code>--de-model</code>	<code>w0</code> (Stage A); <code>cpl</code> (Stage B)	Stabilise then explore CPL
<code>--seed-width-frac</code>	0.30	Prevents sticky starts and NaN τ
<code>--no-kde-move</code>	omit	Allow correlated proposals
<code>--delta-param</code>	omit on BAO-only	Re-enable only after SN/Planck
<code>--noether-strength</code>	1.0–1.5	Avoid penalty-dominated fits
<code>--nwalkers/--nsteps/--nburn</code>	128 / 12000 / 3000	Extend if <code>steps_post/τ_{max} < 20</code>

Changelog

v1.1 — 13 Nov 2025 (AEST)

- Added staged workflow (w0 → CPL → +SN/+Planck → FR extras).
- Guidance for DESI-only mixing: wider seeds, enable correlated moves, limit active params initially.
- Noether strength tuning for BAO-only (1.0–1.5).
- Updated figures section and diagnostics targets (steps_post/ τ_{max}).
- Documented c_0 vs c_{of_z} insensitivity of χ^2_{BAO} at this stage.